Inventory Management System

Senior Software Engineer — Interview Assignment

This document contains the complete technical assignment specification including module requirements, acceptance criteria, evaluation rubric, and UI wireframes.

Tech Stack	Modules	Wireframes
React · Node.js · PostgreSQL	**Tenant · Product Inventory · Order**	**11 Screen Designs**

Note: Wireframes are provided as reference only. Candidates are encouraged to improve on the designs. All ambiguities should be resolved with stated assumptions.

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Technical Deliverables & Evaluation Criteria

Expected Deliverables

Database Schema — Tables for Tenant, Product, Inventory, and Order with proper foreign key / reference relationships.

Backend API — RESTful or GraphQL endpoints supporting CRUD operations with multi-tenant data isolation.

Frontend UI — Responsive pages matching the structural requirements outlined in each module section.

Architectural Flow — Clean separation of concerns: Routes → Controllers → Services → Repositories.

Evaluation Criteria

Criterion	What Is Being Assessed
Multi-Tenancy	Data scoped per tenant. Proper tenant ID propagation. No data leakage across tenants.
Vibe Coding & Speed	Use of modern tooling and AI assistance to rapidly scaffold standard list/detail/form views.
Data Integrity	Rules enforced: orders only map to Active products; insufficient inventory → Pending status.
Code Quality	Reusable components, consistent API response shapes, clean error handling.

Global UI Requirements

Left Navigation	Fixed sidebar with links for Tenant, Product, Inventory, and Order. Active section highlighted.
Top Header	Application logo/name on the left; User Icon/Avatar on the right. No global search bar.
Tenant Scope	Every List page (Product, Inventory, Order) must include a Tenant Dropdown at the top right. Changing the tenant reloads the table showing only that tenant's data.

Entity Mapping

The four core entities are scoped within a Tenant. The flow is:

Tenant → **Product** → **Inventory** → **Order**

An Order references a Product. A Product's stock is tracked via Inventory. All entities are scoped to their parent Tenant.

Module 1 — Tenant

User Intent: As a system administrator, I want to manage different organizational tenants so that data can be segregated appropriately.

User Story 1.1		
As a Admin	**I want** see a list of all tenants	**So that** I can monitor active accounts

Acceptance Criteria:

- A table is displayed with columns: Tenant Name, Tenant ID, Status (Active/Inactive).
- Pagination and basic text search are supported.

User Story 1.2		
As a Admin	**I want** create a new tenant	**So that** a new organization can use the system

Acceptance Criteria:

- Form field: Tenant Name * (required).
- System checks if the Tenant Name already exists; shows inline error if duplicate.
- On valid submission, tenant is created and user is redirected to Tenant List.

Module 2 — Product

User Intent: As a user, I want to define the catalog of items our tenant sells or tracks.

User Story 2.1		
As a User	**I want** view products specific to my selected tenant	**So that** I can browse our catalog

Acceptance Criteria:

- Table columns: Name, SKU, Category, Status (Active/Inactive).
- List supports Search, Pagination, and Sorting.
- No Import/Export functionality.
- Row Action Menu (■) with View, Edit, and Delete options.
- Tenant Dropdown at top right filters table to selected tenant data only.

User Story 2.2		
As a User	**I want** view, edit, or delete a product	**So that** I can keep catalog data accurate

Acceptance Criteria:

- View → Product Details page with Edit and Delete icon buttons.
- Edit form — required fields (*): SKU*, Product Name*, Category*, Reorder Threshold*, Cost per Unit*.
- SKU is read-only after creation.
- Product Status toggle: Active / Inactive (no validation required).

Module 3 — Inventory

User Intent: As a user, I want to manage the inventory levels of my products so we know what is available to fulfill orders.

User Story 3.1		
As a User	**I want** see current inventory levels for all products in my tenant	**So that** I can identify stock positions at a glance

Acceptance Criteria:

- Table columns: Product Name, SKU, Cost per Unit, Current Inventory, Reorder Threshold.
- Row Action Menu (■): View, Edit, Delete.
- Summary tile highlights count of products below the Reorder Threshold.

User Story 3.2		
As a User	**I want** view detailed inventory data and quickly update inventory values	**So that** I can correct stock levels without navigating away

Acceptance Criteria:

- Detail view shows product info, Cost per Unit, and current inventory numbers.
- Current Inventory * (required) input field with 'Update Stock' button for quick updates.
- Product Name on detail page is a hyperlink navigating to the Product Details page.

Module 4 — Order

User Intent: As a user, I want to create and track orders placed against our product inventory.

User Story 4.1		
As a User	**I want** see a list of orders for my tenant	**So that** I can monitor order statuses

Acceptance Criteria:

- Table columns: Order ID, Product, Quantity, Status, Date.
- Row Action Menu (■): View, Edit, Delete.
- Summary tiles showing Total Orders, Total Pending, Total Created counts.

User Story 4.2		
As a User	**I want** create a new order and ensure inventory is available	**So that** I can validate stock before placing an order

Acceptance Criteria:

- Select Product * dropdown shows ONLY Active products (Inactive products are excluded).
- Requested Quantity * is a required field.
- If Inventory ≥ Requested Quantity → order saved with status Created.
- If Inventory < Requested Quantity → order saved with status Pending.

User Story 4.3		
As a User	**I want** view order details and process confirmations	**So that** I can act on orders from a single screen

Acceptance Criteria:

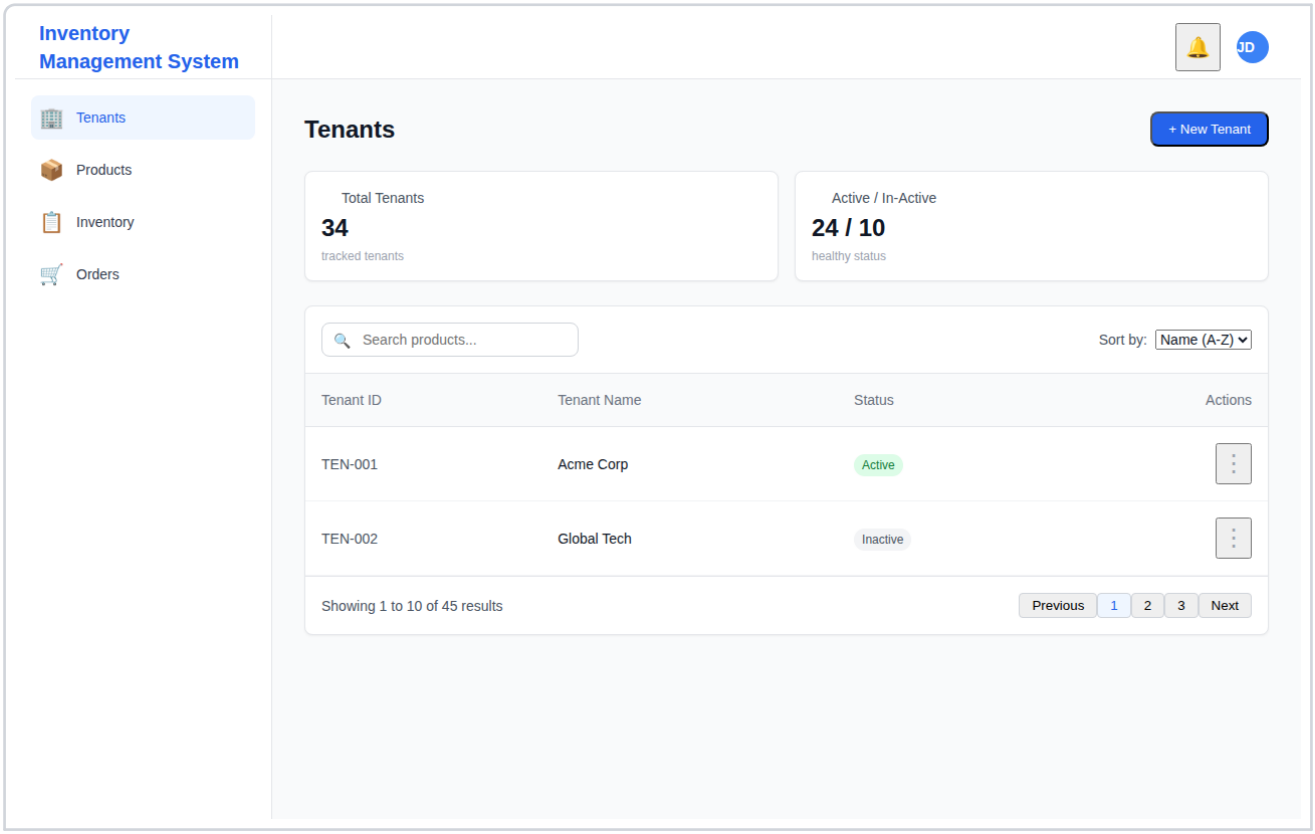
- Detail view shows: product info, requested quantity, current inventory data.
- Primary action buttons: Confirm and Cancel.
- Product name on detail page links to the Product Details page.

UI Wireframes — Screen Reference

The following screens are provided as reference wireframes. Candidates are welcome to improve the layout and visual design, provided all functional requirements are met.

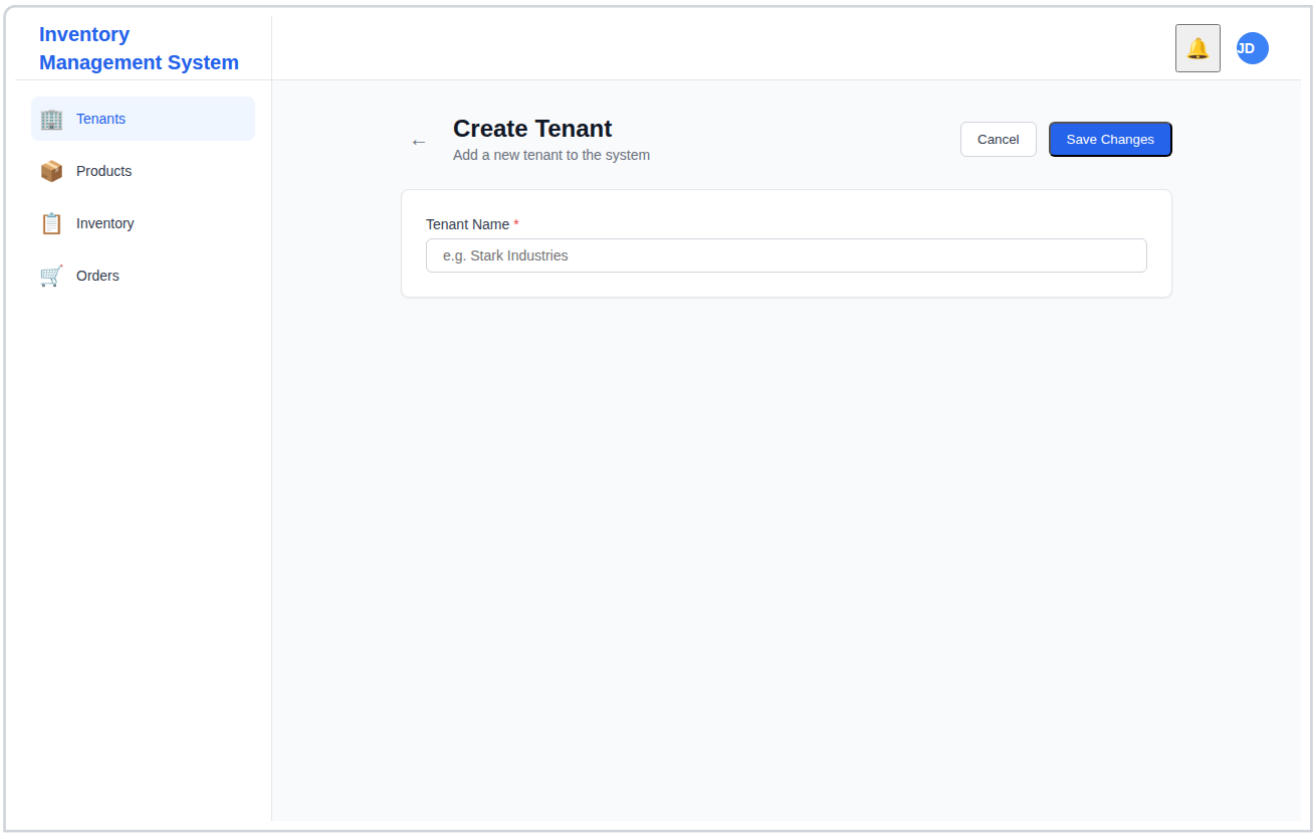
Tenant Module

Tenant List



Wireframe: Tenant List

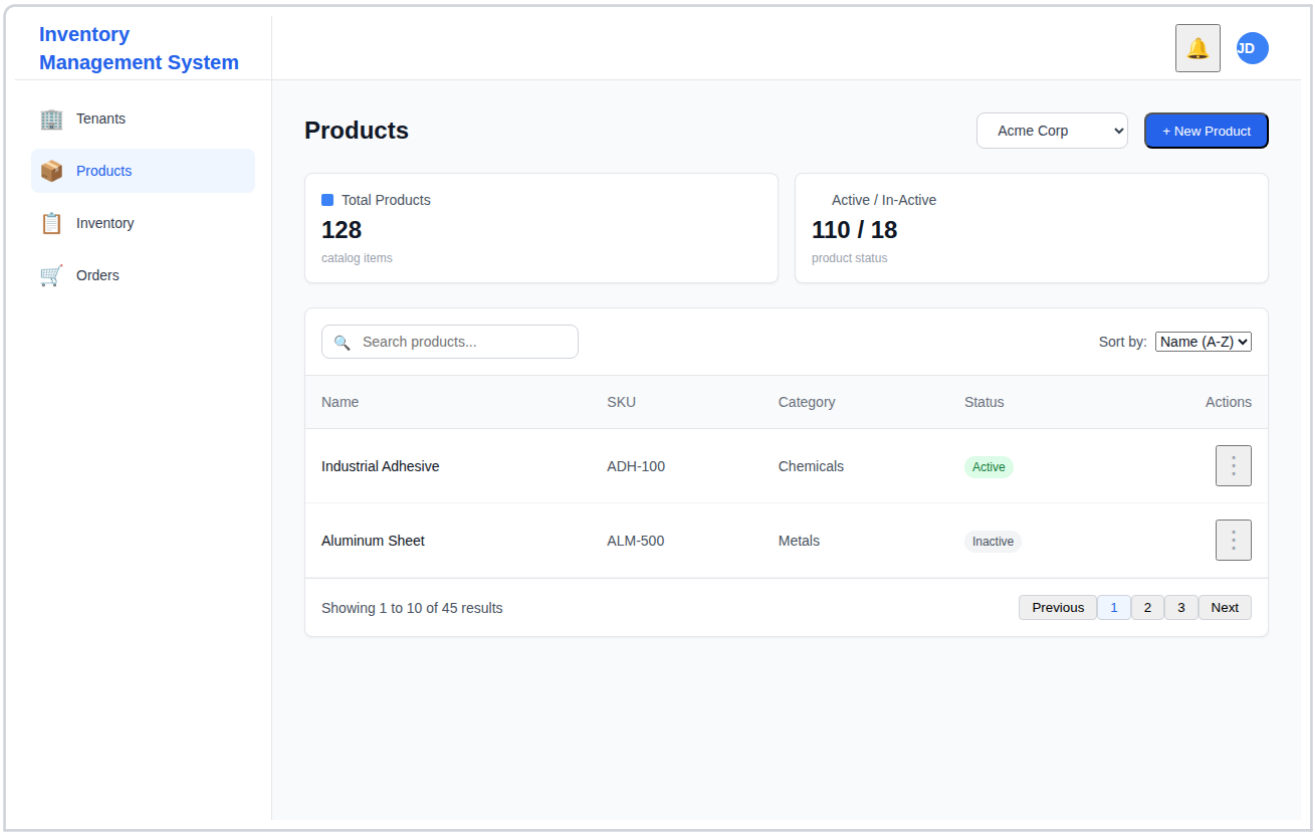
Tenant – Create/Edit



Wireframe: Tenant – Create/Edit

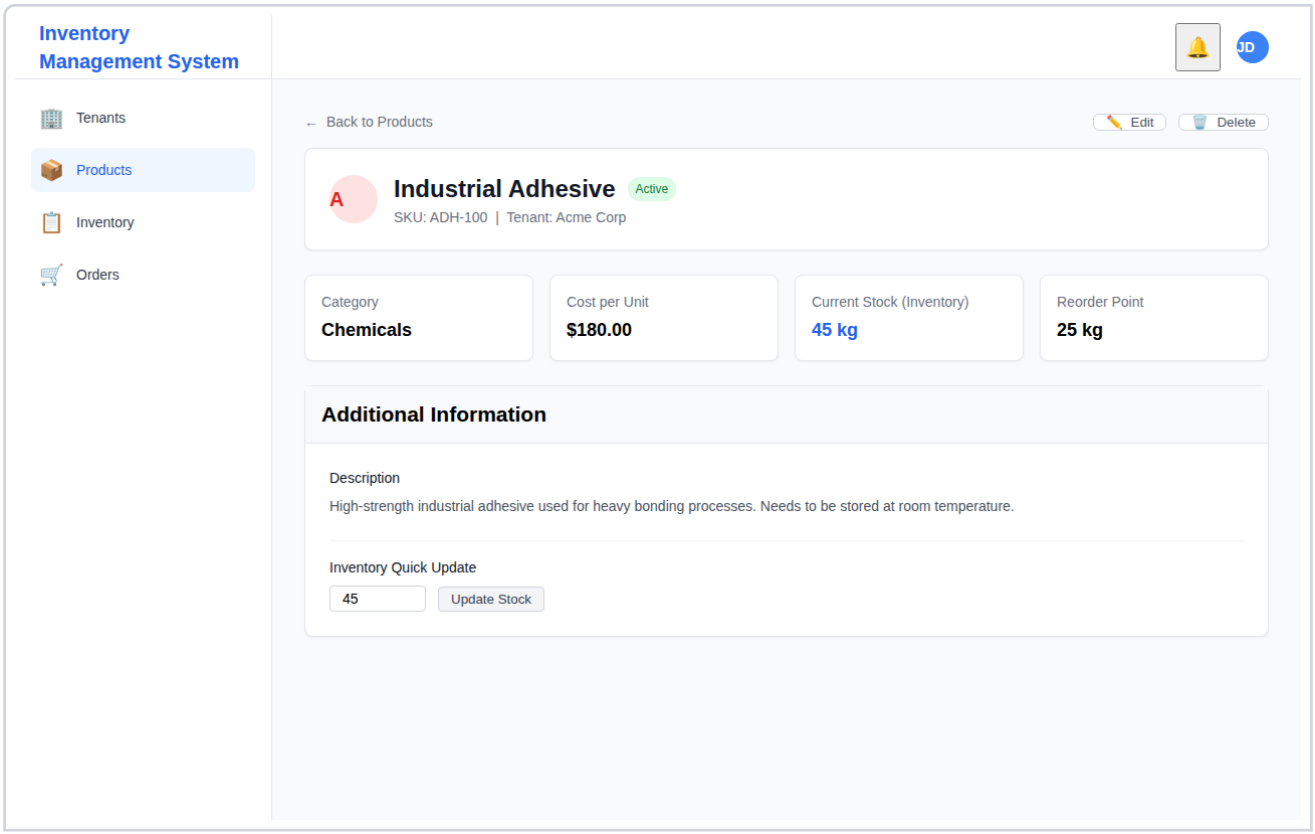
Product Module

Product List



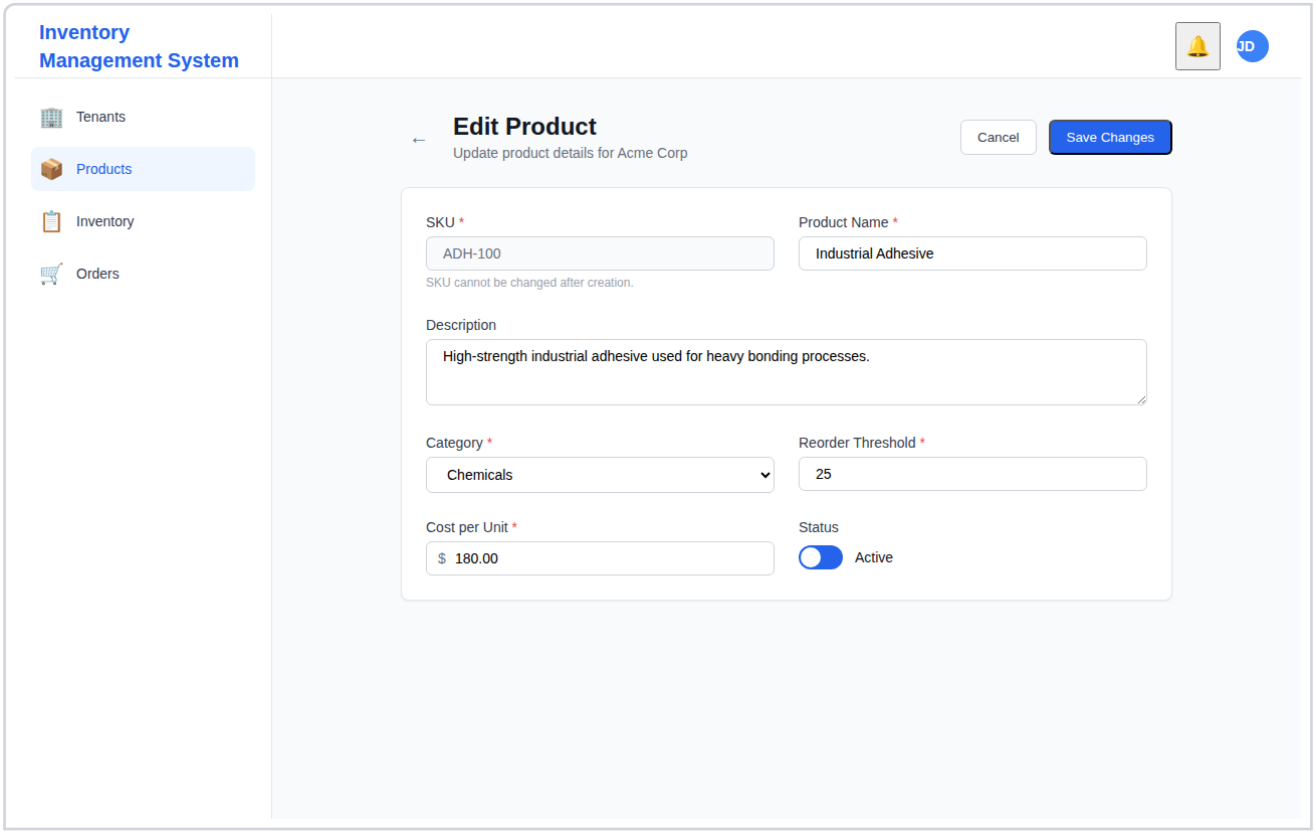
Wireframe: Product List

Product Detail



Wireframe: Product Detail

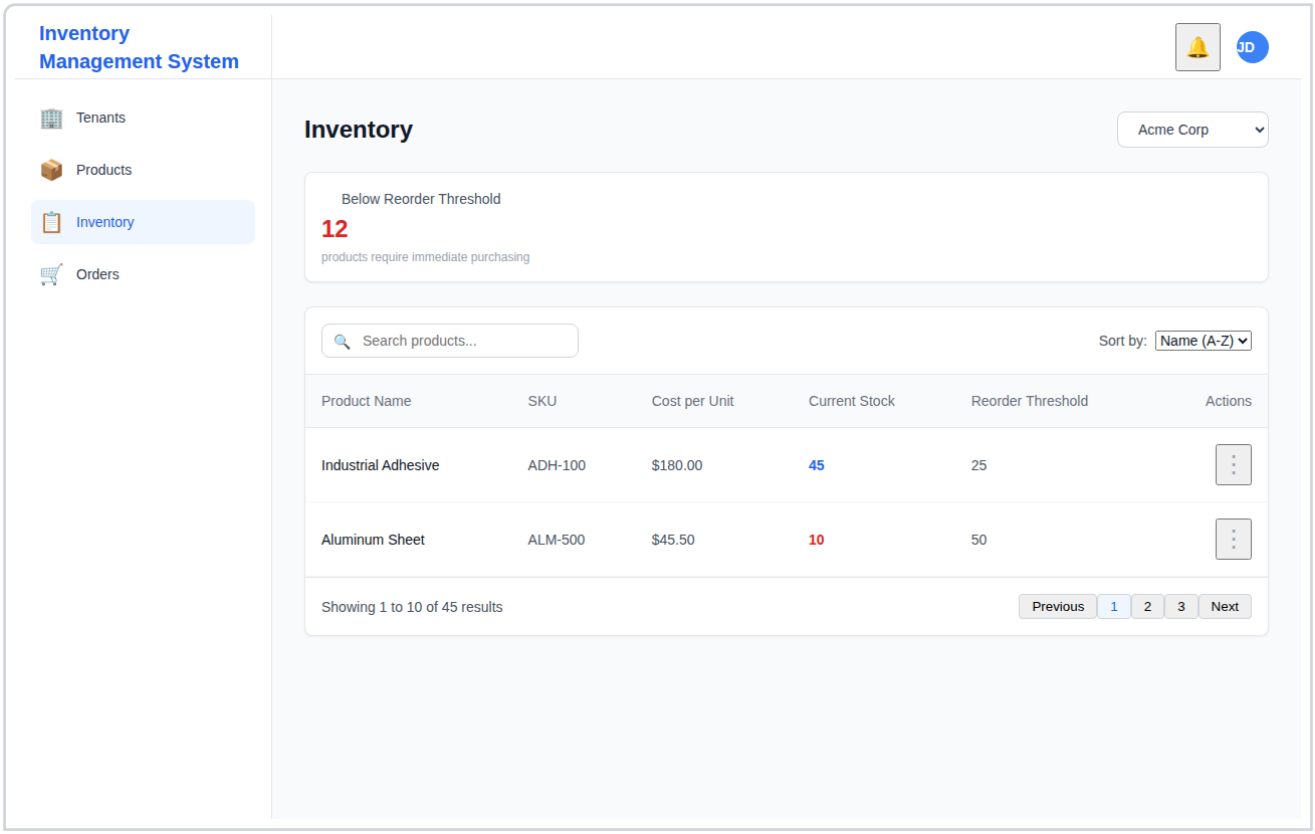
Product Edit



Wireframe: Product Edit

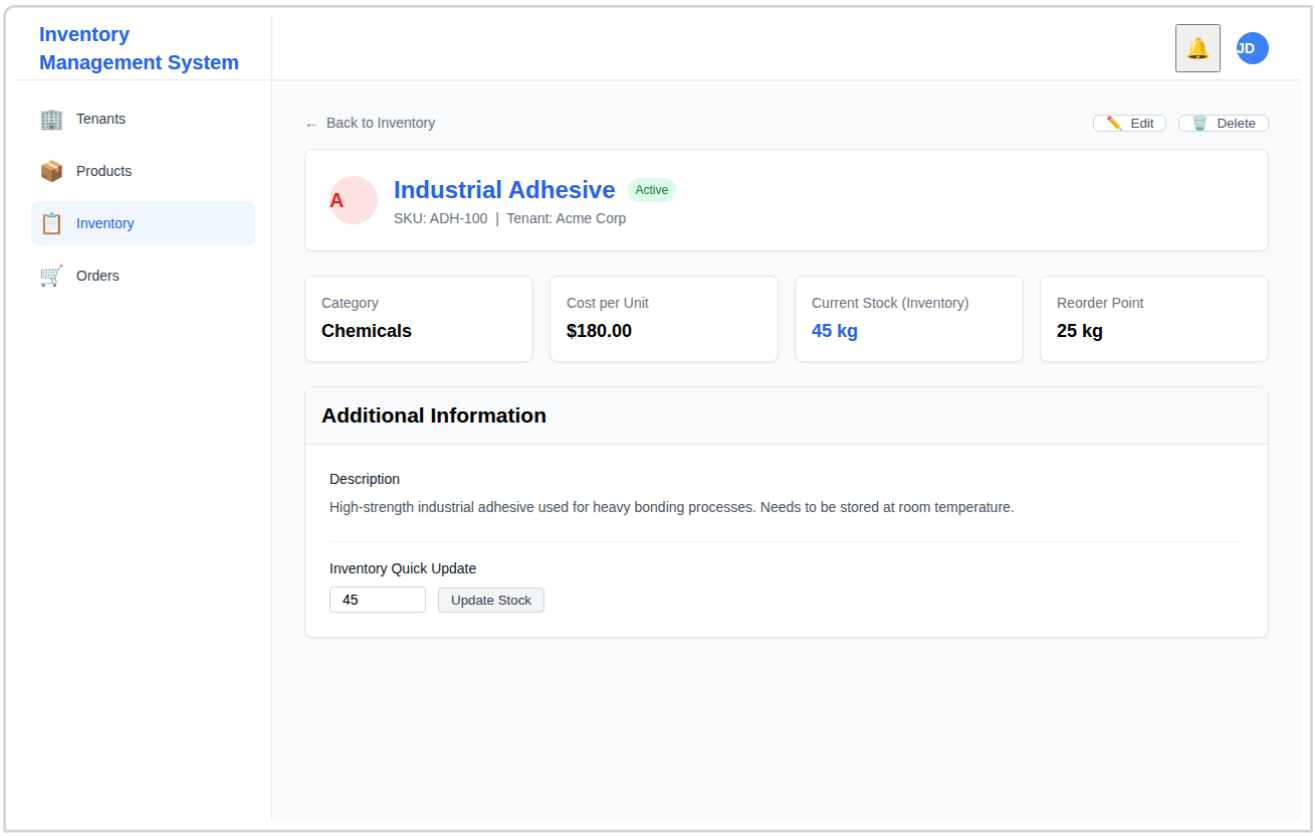
Inventory Module

Inventory List



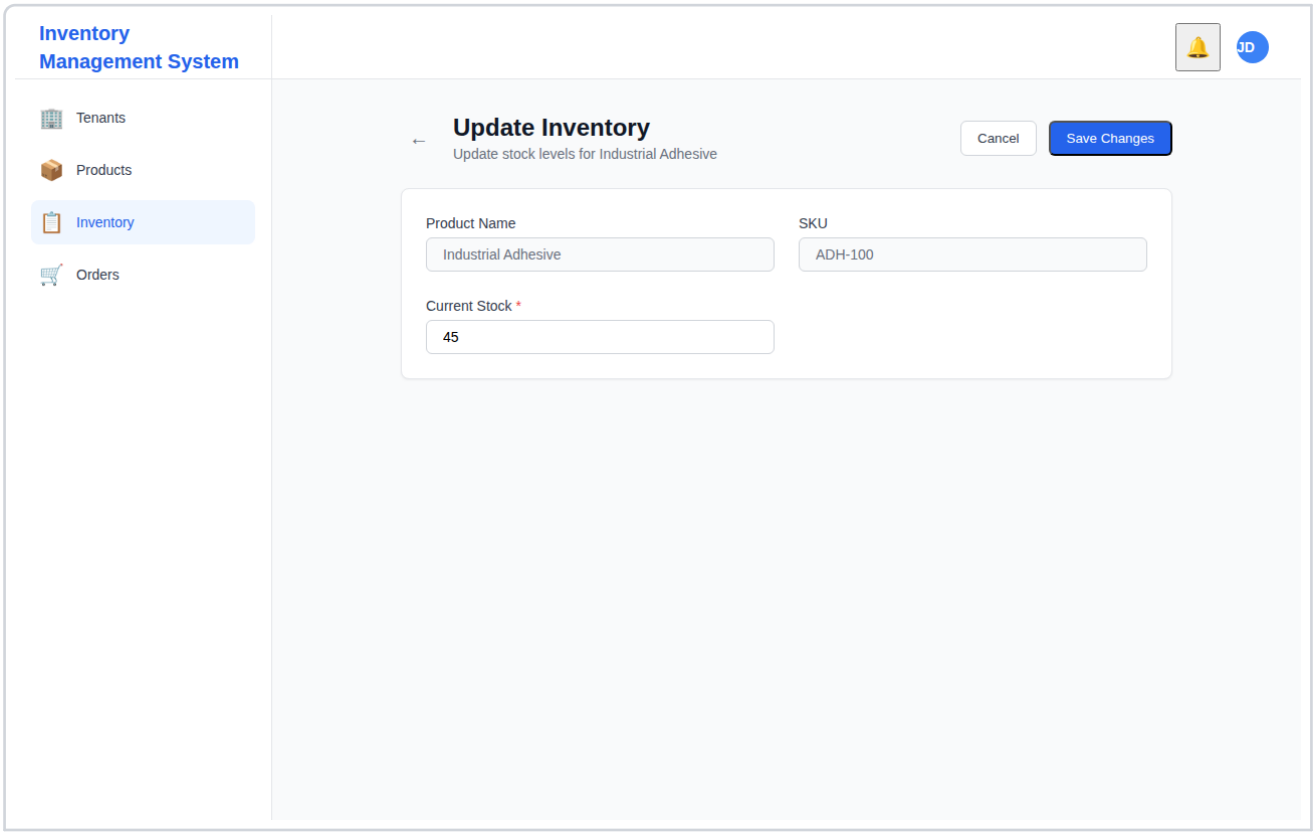
Wireframe: Inventory List

Inventory Detail



Wireframe: Inventory Detail

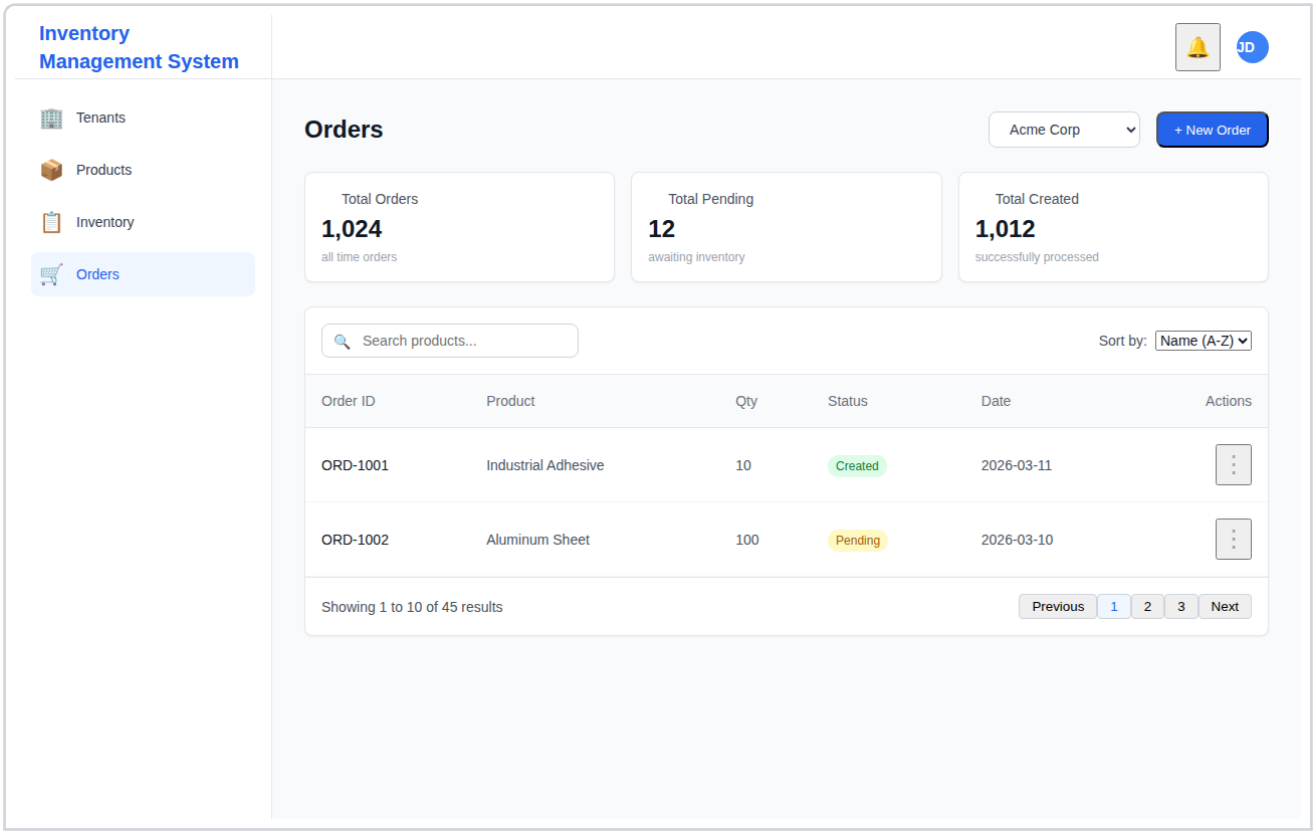
Inventory Edit



Wireframe: Inventory Edit

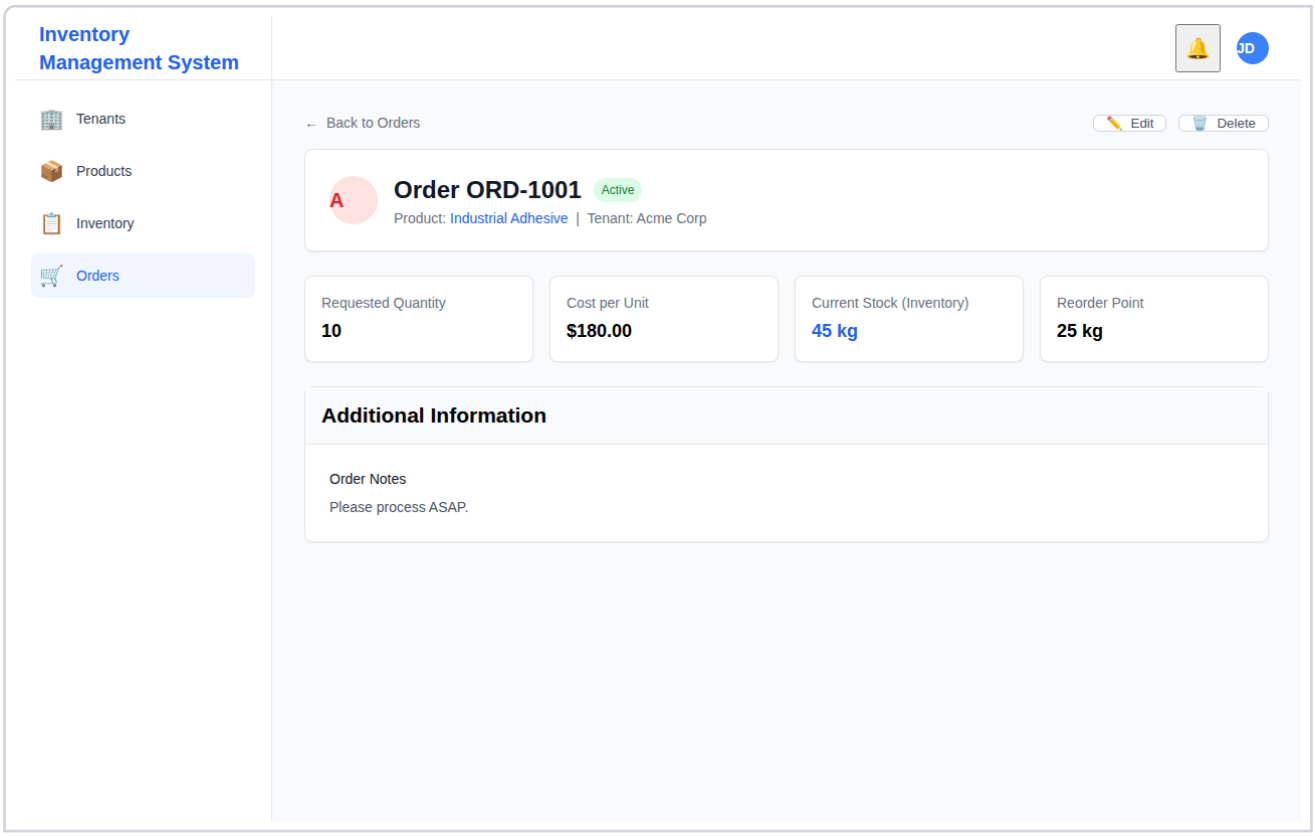
Order Module

Order List



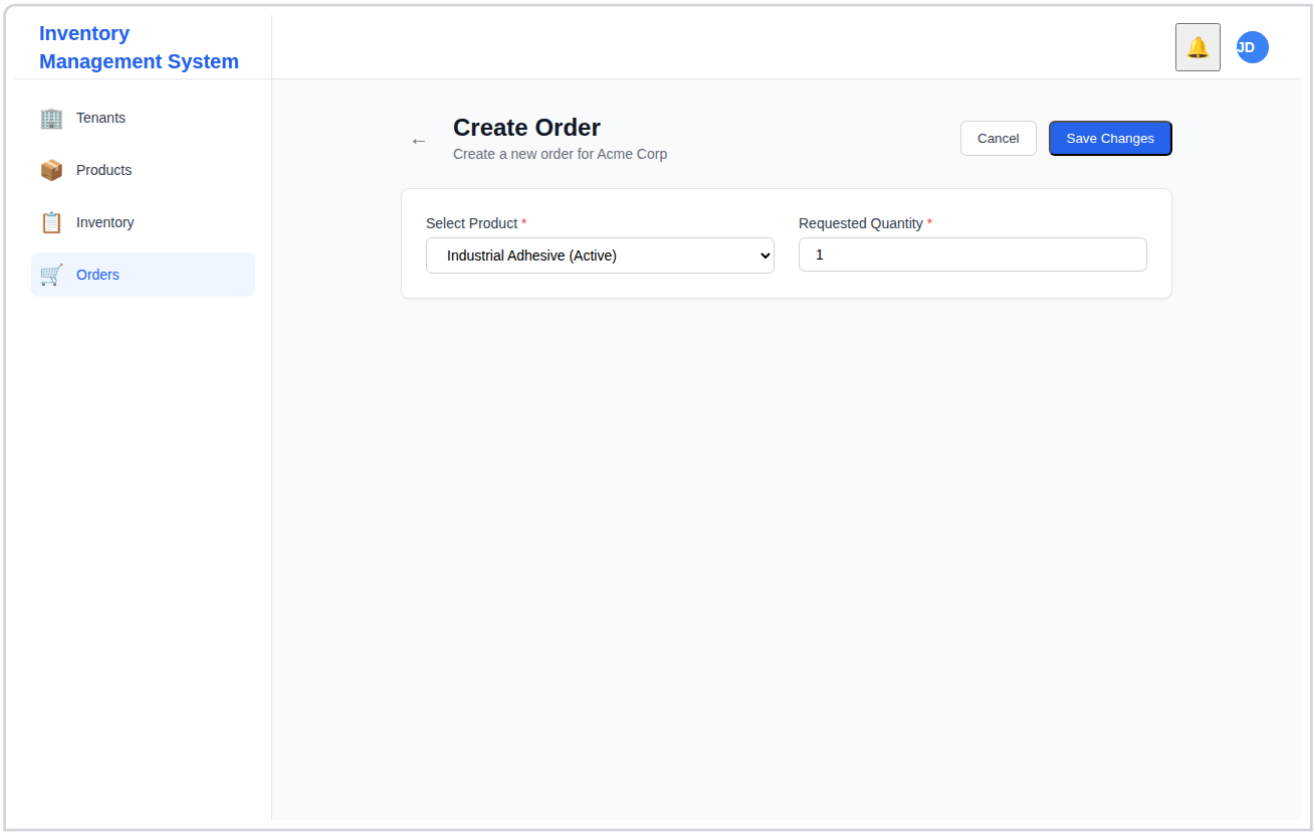
Wireframe: Order List

Order Detail



Wireframe: Order Detail

Order Create/Edit



Wireframe: Order Create/Edit

■ Assumptions & Creativity

If you encounter any gaps or ambiguities in these requirements, please define your own assumptions and proceed with development. You will have the opportunity to discuss and highlight these assumptions during the interview. You are also welcome to bring your own creativity and innovation to this application. The wireframes are for reference only — a better UX is always appreciated.

Upon completion, share your solution as a public GitHub repository link.